

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Science in Information Technology (Msc. (I.T))

Semester-II University Examination May - 2018

MS 210 Wireless Communications

Date: 07.05.18, Monday Time: 10.00 a.m. to 01.00 p.m.

Maximum Marks: 70

Instructions:

- 1. Attempt both sections in separate answer books.**
- 2. Figure to right indicates marks.**
- 3. Make suitable assumption when necessary.**

SECTION - I

Q1. (A) Answer Following. [05]

1. _____ signals are analog or digital signals that repeat pattern over time.
2. A capacity of cell can be increased by _____.
3. What is shape of cell in cellular system?
4. A _____ antenna is used in satellite communication
5. What are different components of a cell?

Q1. (B) Do as Directed. [06]

1. What is a signal? What are different types of signals?
2. Give full form of: 1) BS 2) MTSO.
3. Define following: 1) Beam Width 2) Reception Pattern

Q2. Answer Following. [12]

1. What is multiplexing? Differentiate between: FDM and TDM
2. How can we increase capacity of a cell? Discuss different techniques for increasing cell capacity.

OR

Q2. Answer Following. [12]

1. What is communication system? List down its components.
2. How does a cellular system work? Explain in detail.

Q3. Answer Following. [12]

1. What is handoff? What are different parameters of handoff? Explain each parameter in 1-2 lines.
2. Write short note on different types of propagation modes.

OR

Q3. Answer Following. [12]

1. Apart from main function of managing call, what are the other functions of an MTSO?
2. List down different categories of noise and give example for each.

SECTION - II

Q4. (A) Answer Following.

1. Which encoding technique is based on sampling theorem? **[01]**
2. Define following: 1) Data Element 2) Data Rate 3) Flow Control. **[03]**
3. What determines how successful a receiver will be in interpreting an incoming signal? **[01]**

Q4. (B) Do as Directed.

[06]

1. Differentiate between analog and digital signals.
2. Give full form of: 1) ASK 2) PSK.
3. What is parity check method? Give example of odd and even parity.

Q5. Answer Following.

[12]

1. List down and explain different signal encoding criterion in brief.
2. Explain free space loss with example.

OR

Q5. Answer Following.

[12]

1. Write short note on PCM and DM.
2. Write short note on different types of antenna with their applications.

Q6. Answer Following.

[12]

1. Write short note on Wi-Fi.
2. Explain wireless ad-hoc network in detail.

OR

Q6. Answer Following.

[12]

1. Explain CRC technique with example.
2. Use ASK FSK and PSK techniques on following data to generate appropriate modulation chart: 00110100010.